

Olaparib in Breast Cancer (All Subtypes)

Drug Summary

Olaparib (Lynparza) is a first-in-class PARP inhibitor used in breast oncology for tumors with defective DNA repair, especially those harboring **germline BRCA1/2 mutations** ¹ ². It is FDA-approved for **HER2-negative** breast cancers in both **metastatic** and **early-stage (adjuvant)** settings, contingent on a **deleterious BRCA mutation** ¹ ². In metastatic disease, olaparib monotherapy improved progression-free survival in patients with BRCA-mutated, HER2-negative tumors (including both triple-negative and ER-positive subtypes) after prior chemotherapy ³. In the early-stage, high-risk setting, 1 year of adjuvant olaparib significantly reduced invasive recurrence and improved overall survival in **gBRCA1/2-mutated, HER2-negative** patients following standard neoadjuvant/adjuvant therapy ⁴ ⁵. These benefits are most pronounced in “**BRCAness**” contexts – e.g. triple-negative breast cancers (TNBC) or certain luminal B cancers with homologous recombination deficiency (HRD) – where olaparib exploits synthetic lethality ⁶ ⁷.

Mechanistically, olaparib **inhibits PARP1/2 enzymes** involved in single-strand DNA break repair, leading to accumulation of DNA damage and tumor cell death, especially in BRCA/HRD-positive cancers that cannot perform error-free homologous recombination repair ⁶ ⁸. It has become an integral targeted therapy in BRCA-mutated breast cancer: current guidelines (NCCN, ASCO) recommend germline BRCA testing for patients with high-risk or metastatic breast cancer, and endorse olaparib for **gBRCA1/2-mutant TNBC or HR-positive/HER2-negative cancers** after standard treatment ⁹ ¹⁰. Olaparib is also being investigated in combination with other therapies (e.g. immune checkpoint inhibitors, DNA damage response inhibitors) to extend its benefit beyond BRCA-mutant populations, reflecting its emerging role across molecular subtypes of breast cancer with DNA repair vulnerabilities.

Identifiers & Synonyms

- **ChEMBL ID:** ChEMBL521686 ¹¹
- **DrugBank ID:** DB09074 ¹² ¹³
- **Synonyms/Trade Names:**
- **Lynparza** – Brand name for olaparib ¹⁴
- **AZD-2281 (AZD2281)** – AstraZeneca investigational code for olaparib ¹⁵
- **KU-0059436** – Early research code for olaparib (KuDOS Pharmaceuticals) ¹⁵
- **MK-7339** – Merck code for olaparib in clinical trials ¹⁶ ¹⁷

Mechanism of Action (Breast Cancer Context)

Olaparib is a **potent inhibitor of PARP1 and PARP2** (poly ADP-ribose polymerases) ¹⁸, enzymes that detect single-strand DNA breaks and initiate base-excision repair. In breast cancer cells with intact DNA repair, PARP inhibition leads to accumulation of single-strand breaks that convert into double-strand breaks (DSBs) during replication ¹⁹. **BRCA1/2 and other homologous recombination (HR) proteins** (e.g. PALB2,

RAD51) are normally required to faithfully repair such DSBs ²⁰ ²¹ . Tumors with **HR deficiencies (HRD)** – often due to germline **BRCA1/2 mutations**, but also possible through somatic loss or epigenetic silencing of HR genes – cannot repair these lesions, leading to **genomic instability and cell death** when exposed to olaparib ⁶ ⁸ . This phenomenon exemplifies **synthetic lethality**, wherein PARP inhibition selectively kills HR-deficient cancer cells while sparing normal cells with functional HR ²² ⁷ .

In BRCA-mutated **triple-negative breast cancers**, which inherently lack HR repair, olaparib induces profound DNA damage, resulting in tumor cell apoptosis and growth arrest ⁶ ³ . HR-deficient cells also experience stalled replication forks; olaparib “traps” PARP1 on DNA, preventing repair and exacerbating replication stress ⁸ ²³ . Conversely, HR-proficient breast cancers can often survive PARP inhibition by repairing DSBs via HR, which explains why sporadic (non-BRCA) tumors are less sensitive ⁷ . Notably, **hormone receptor-positive (HR+) HER2-negative** breast cancers with BRCA2 mutations derive benefit from olaparib as well ⁹ , although in HR+ patients wild-type for BRCA, HR repair remains intact and PARP inhibitors alone show limited efficacy. Olaparib’s mechanism thus fits into breast cancer treatment algorithms as a targeted therapy for the **DNA repair-defective subset** of tumors. It is often used after chemotherapy (and endocrine therapy if ER-positive) in those with BRCA mutations ⁹ , and as adjuvant therapy to eradicate minimal residual disease in high-risk, BRCA-mutant early breast cancer ² ¹⁰ . Additionally, PARP inhibition may activate innate immunity (cGAS-STING pathway) by increasing cytosolic DNA, providing a rationale for combining olaparib with immunotherapy in TNBC ²⁴ ²⁵ . In summary, olaparib’s MoA centers on exploiting defective HR DNA repair – common in basal-like/TNBC and some luminal B tumors – to induce lethal DNA damage accumulation.

Primary Targets (Human)

Olaparib’s primary molecular targets in human cells are the **PARP (poly[ADP-ribose] polymerase) enzymes** involved in DNA single-strand break repair:

- **PARP1 (ARTD1):** Nuclear enzyme that senses DNA single-strand breaks and recruits repair proteins via PARylation. Olaparib binds to PARP1’s catalytic domain, **inhibiting its activity and “trapping” PARP1 on DNA**, which prevents repair of nicks ¹⁸ ²³ . This is the main target driving cytotoxicity in HR-deficient cells.
- **PARP2:** A partner enzyme to PARP1 in base excision repair. Olaparib also inhibits PARP2 ¹⁸ , contributing to the accumulation of unrepaired DNA lesions.
- **PARP3:** Olaparib has inhibitory activity against PARP3 as well ¹⁸ , though PARP3’s role in the cytotoxic mechanism is less prominent than PARP1/2.

These targets have been confirmed through biochemical assays and DrugBank annotations, which classify olaparib as a **PARP1/2 (and PARP3) inhibitor** ²⁶ ¹⁸ . By inhibiting PARP1/2, olaparib effectively cripples the base-excision repair pathway in tumor cells, leading to lethal DNA damage if HR repair is absent. No other human proteins are intentionally targeted by olaparib at therapeutic concentrations (though off-target effects on other PARP family members or transporters may occur at higher doses). The primary **HGNC gene symbols** for olaparib’s targets are **PARP1** and **PARP2** (with PARP3 as a minor target), validated by binding assays and cellular studies ²⁶ .

Pathways (Overview)

Olaparib perturbs multiple cellular pathways in breast cancer, chiefly those related to DNA damage sensing, repair, and cell survival. Key pathways and gene sets affected include:

- **Homologous Recombination DNA Repair (HRR)** – *MSigDB Hallmark DNA Repair* and *Reactome Homologous Recombination*. Olaparib's efficacy is mechanistically linked to HRR dysfunction; HR-deficient pathways (BRCA1/2, PALB2, etc.) are central to its synthetic lethal effect ⁶ ⁸. Upregulation or restoration of HRR confers **resistance**, while HRR downregulation (HRD) confers **sensitivity** to olaparib (see details below). This pathway is core to olaparib's **MoA**.
- **Base Excision Repair (BER) / Single-Strand Break Repair** – *KEGG base excision repair (hsa03410)*. Olaparib directly inhibits the BER pathway by blocking PARP1/2. In breast cancer cells, suppression of BER leads to accumulation of DNA single-strand breaks, linking this **mechanistic pathway** to downstream lethal damage in HRD tumors ¹⁸ ²³.
- **Cell Cycle Checkpoints (G2/M, S-phase)** – e.g. *Reactome ATR/CHK1 activation* and *Hallmark G2M Checkpoint*. Olaparib-induced DNA damage activates ATR/ATM-mediated checkpoints. If these checkpoints are intact, cells may arrest and attempt repair; if abrogated (e.g. ATM mutation), cells proceed to mitosis with damage, enhancing cytotoxicity ⁶. Thus checkpoint pathways modulate sensitivity: loss of ATM/ATR checkpoint can increase olaparib sensitivity (by preventing DNA repair) ⁶, whereas overactive ATR/CHK1 signaling can promote resistance by stabilizing replication forks ²⁷. This is both a **sensitivity-associated** and **resistance-associated** pathway depending on context.
- **Replication Fork Protection** – includes factors like 53BP1/Shieldin, ATR, and fork remodeling proteins. This pathway isn't a single canonical gene set, but conceptually, **fork stability mechanisms** (e.g. 53BP1-RIF1 pathway, EZH2-mediated fork protection) profoundly influence olaparib response ²⁸ ²⁷. Upregulation of fork protection (e.g. via loss of 53BP1 or activation of ATR) allows tumor cells to survive PARP inhibition (**resistance-associated**), whereas inability to protect forks (e.g. **SLFN11** presence forcing fork collapse) correlates with sensitivity ²⁹ ³⁰.
- **PI3K/AKT/mTOR Signaling** – *MSigDB Hallmark PI3K_AKT_MTOR_SIGNALING*. This survival pathway interacts with DNA repair networks. Hyperactivation of PI3K/AKT can promote HR repair and cell survival under genotoxic stress ³¹, emerging as a **bypass resistance pathway** in BRCA-mutated cancers. Indeed, PARP inhibitor exposure can induce pro-survival AKT signaling, limiting apoptosis ³². Conversely, inhibiting PI3K/AKT downregulates BRCA1/2 expression and induces HRD, **sensitizing** tumors to olaparib ³¹ ³³. Thus, this pathway has both mechanistic and resistance implications, and combination trials of olaparib with AKT or mTOR inhibitors have shown synergistic activity in breast cancer models ³⁴.
- **ER Signaling (Hormone Receptor Pathway)** – *Hallmark Estrogen Response (Early/Late)*. While not directly targeted by olaparib, ER signaling is relevant in **HR+/HER2-** breast cancers. BRCA2-mutant ER+ tumors benefit from olaparib ⁹, but wild-type ER+ tumors (with intact HR) generally rely on endocrine therapy. There is no evidence that estrogen receptor signaling per se is altered by olaparib; instead, ER pathway status helps define which patients qualify (HR+ patients must have

failed endocrine therapy before PARP inhibitor use ⁹). Thus, this pathway is **subtype-specific** context: HR+ patients respond to olaparib only if an HRD biomarker (like BRCA mutation) is present.

- **HER2/ERBB Signaling** – e.g. *Reactome Signaling by ERBB2*. HER2-positive breast cancers were excluded from pivotal olaparib trials ³⁵ ; accordingly, olaparib's role in HER2-driven pathways is minimal. Preclinically, HER2 and PI3K pathways crosstalk with DNA repair, but clinically **HER2+ patients are treated with anti-HER2 agents** rather than PARP inhibitors unless they also carry BRCA mutations. This pathway is noted for completeness, but is not a major focus of olaparib's mechanism (i.e. olaparib is not indicated for HER2+ tumors without HRD).
- **Immune Response and Tumor Microenvironment (TME)** – *MSigDB Hallmark Interferon Alpha/Gamma Response; Immune Sig: T-cell infiltration*. PARP inhibition can activate innate immune signaling by causing cytosolic DNA accumulation. Notably, **cGAS-STING pathway** activation by olaparib induces type I IFNs and chemokines, promoting **T-cell recruitment** ²⁴ ³⁶ . Immune-related pathways (e.g. interferon signaling, antigen presentation) have been found upregulated in BRCA-mutant models treated with olaparib, especially when combined with STING agonists ³⁷ ³⁸ . A **“hot” immune microenvironment** (e.g. pre-existing T cell infiltration in TNBC) may augment olaparib's efficacy, whereas immune-evasive tumors could be less responsive (leading to trials combining olaparib with PD-1/PD-L1 inhibitors). Thus, immune pathways are an **adaptive mechanistic aspect** of olaparib action, particularly in TNBC: PARP inhibition can convert immunologically “cold” tumors into “hotter” ones via STING-mediated inflammation ²⁴ ²⁵ . This is being leveraged in ongoing trials (e.g. olaparib + checkpoint inhibitor).
- **Apoptosis Pathway** – *Hallmark Apoptosis; p53 Pathway*. The ultimate fate of olaparib-treated cancer cells is apoptosis or mitotic cell death. Pro-apoptotic signaling (e.g. functional p53 pathway, BIM activation) enhances cell killing by olaparib, whereas defects in apoptosis (e.g. loss of p53 or overexpression of BCL-2) could allow cells to survive longer. Many BRCA1-mutant TNBCs also have TP53 mutations, yet they still respond well to PARP inhibition via mitotic catastrophe rather than classic p53-mediated apoptosis. Nonetheless, **intact apoptotic machinery** is generally a favorable factor – for example, cells with wild-type p53 can undergo apoptosis in response to PARP inhibitor-induced DNA damage, contributing to drug sensitivity ⁶ . In contrast, p53-deficient cancers might rely on alternative death pathways. Thus, apoptosis regulation is a **downstream mechanistic pathway** affecting response.
- **Drug Efflux and Transporter Pathways** – e.g. *Reactome ABC-family transporters*. Upregulation of multidrug resistance transporters (especially **ABCB1/P-glycoprotein**) in tumor cells can significantly impact olaparib efficacy ³⁹ . These transporters pump olaparib out of cells, reducing intracellular drug levels. While not part of olaparib's intended mechanism, **ABC transporter pathways** are relevant as a **resistance mechanism**: ABCB1 overexpression has been observed in PARP inhibitor-resistant breast and ovarian cancer models ³⁹ ⁴⁰ . Conversely, inhibiting or avoiding P-gp efflux can restore sensitivity. This pathway highlights the intersection of drug metabolism with resistance.
- **EMT/Metastasis and Other pathways**: Epithelial–mesenchymal transition (EMT) and related pathways (TGF- β signaling, etc.) have not been conclusively tied to olaparib response, but more mesenchymal, genomically unstable tumors (like basal/TNBC) are often the target population. Metabolic pathways (NAD⁺ salvage, for example) are involved in PARP function (PARP consumes NAD⁺), but therapeutic modulation of metabolism in conjunction with olaparib is still experimental.

Finally, angiogenesis or cytokine networks may indirectly influence response (e.g. via tumor perfusion or immune cell trafficking) but are not primary drivers of sensitivity/resistance in current evidence.

In summary, olaparib in breast cancer predominantly engages DNA damage repair pathways (HR, BER) and elicits cell-cycle arrest and apoptosis, with secondary effects on immune signaling. Pathways that **enhance DNA repair or cell survival** (HR restoration, fork protection, PI3K, drug efflux) tend to cause **resistance**, whereas those that **impair DNA repair or enhance cell death** (HRD, checkpoint loss, immune activation) are associated with **sensitivity**. The detailed sections below stratify these pathways by upregulation vs downregulation and their impact on olaparib sensitivity or resistance, with attention to breast cancer subtype contexts.

Upregulated Pathways (by context/subtype)

Certain pathways become transcriptionally or functionally **upregulated in breast tumors in response to olaparib exposure or in association with resistance**. Depending on the pathway, this upregulation can either be a compensatory resistance mechanism or part of a favorable anti-tumor response:

- **DNA Repair & HR Restoration:** In tumors that develop resistance, compensatory upregulation of DNA repair pathways is observed. For instance, resistant BRCA-mutant breast cancers can **upregulate homologous recombination activity** via secondary BRCA mutations (gene reversion) or loss of HR suppressors (53BP1) ^{28 41}. This effectively “restores” the HR pathway, allowing efficient DSB repair despite PARP inhibition. Similarly, **increased expression of RAD51 and other HR proteins** has been noted in PARP inhibitor-resistant models, reflecting upregulation of HR repair capacity (a clear resistance pathway in BRCA-deficient contexts). In summary, **reactivation of HR repair** is an upregulated pathway in non-responders, conferring resistance by fixing DNA breaks that would otherwise be lethal. By contrast, **upregulation of error-prone repair** (e.g. alternative end-joining) is less beneficial to the cell and is not commonly reported as a resistance mechanism.
- **ATR/CHK1 and Replication Stress Response:** Upon olaparib treatment, breast cancer cells often activate the ATR-CHK1 checkpoint pathway as a stress response. **ATR signaling upregulation** helps stabilize stalled replication forks and delays cell-cycle progression, giving cells time to repair DNA. In TNBC models, **high ATR/CHK1 activity** has been linked to reduced olaparib efficacy, since it attenuates replication catastrophe ²⁷. Clinically, trials are investigating ATR inhibitors to counteract this feedback. Thus, **upregulation of ATR/CHK1 pathway** is seen in tumors under PARP inhibitor pressure (adaptive response), generally contributing to drug resistance. Consistently, resistant tumors may show increased phospho-Chk1 and other markers of active checkpoint signaling. (Conversely, HRD tumors with ATR pathway defects respond better, as discussed later.)
- **PI3K/AKT Survival Pathway:** PARP inhibition can unintentionally trigger pro-survival signaling. Studies indicate **PARP inhibitor exposure activates AKT** in some cancer cells ³². In breast cancer, especially **HR+/HER2- tumors with PIK3CA mutations**, the PI3K pathway is frequently active. **Upregulation of PI3K/AKT signaling** has been observed as a mechanism of **apoptosis resistance**: activated AKT can prevent apoptotic cell death and may promote DNA double-strand break repair via HR facilitation ³¹. Indeed, in BRCA-mutant TNBC models, PI3K pathway activation (e.g. via PTEN loss) can confer PARP inhibitor resistance unless PI3K is co-inhibited ³³. Clinically, this is relevant as combining AKT or mTOR inhibitors with olaparib is being tested to thwart this survival upregulation

³⁴ . In summary, **upregulation of the PI3K/AKT/mTOR pathway** in olaparib-treated tumors (any subtype) tends to be a **resistance-associated** adaptation, helping tumor cells survive DNA damage. Luminal B breast cancers often have PI3K activation and may be intrinsically less sensitive to PARP inhibitors unless this pathway is curtailed.

- **Immune Activation Pathways:** Interestingly, some pathway upregulation indicates a favorable response. **Type I interferon and STING pathway upregulation** can occur after olaparib treatment, especially in TNBC. This stems from accumulating DNA fragments activating the **cGAS-STING** innate immune sensor ²⁴ . As a result, **interferon-stimulated gene sets** and antigen presentation pathways become upregulated, converting “cold” tumors into more immunogenic ones ³⁷ ³⁸ . For example, in BRCA-mutant TNBC models, olaparib monotherapy led to increased T-cell chemokines and PD-L1 expression (signs of immune activation) ²⁴ . Such **immune pathway upregulation** is associated with **greater sensitivity** or improved outcomes when combined with immunotherapy. Clinically, this is subtype-specific: **immune-rich TNBCs** see the greatest immune pathway induction, whereas luminal tumors (with less T-cell presence) show minimal immune upregulation. Thus, **upregulation of interferon/T-cell infiltration pathways** is a context-dependent phenomenon that can be beneficial (sensitizing tumors to immunologic clearance).
- **Drug Efflux Transporters:** On the negative side, an often-seen adaptive change is **upregulation of ATP-binding cassette (ABC) transporters** such as **ABCB1 (P-gp)** in tumors exposed to olaparib. This is a general stress response where cancer cells pump out xenobiotics. In breast cancer cell lines selected for PARP inhibitor resistance, ABCB1 was markedly overexpressed (20-fold in one study) ⁴² . This **upregulated efflux pathway** reduces intracellular olaparib concentration and is a known resistance mechanism (not specific to a breast subtype, though more commonly reported in triple-negative and ovarian contexts) ³⁹ . Notably, this can co-occur with taxane resistance (since P-gp affects many drugs) ⁴³ . ABC transporter upregulation is a hurdle in maintaining therapeutic drug levels, and inhibitors of P-gp (tariquidar, etc.) can reverse this resistance in preclinical models ³⁹ . While routine clinical testing for ABCB1 is not done, this mechanism is suspected in patients who progress quickly on PARP inhibitors.
- **MYC and HR Gene Expression:** High proliferation and MYC-driven transcriptional programs may also be upregulated in some resistant tumors. MYC can drive expression of HR and replication stress tolerance genes. For example, some PARP inhibitor-resistant TNBC derivatives show upregulation of E2F/MYC target genes that promote cell cycle progression despite DNA damage. This is more speculative, but it underscores that **upregulation of cell proliferation pathways** (through MYC/E2F) can allow tumors to bypass PARP-induced cell cycle arrest, continuing to divide (which might be lethal, but if coupled with other resistance like HR restoration, the cells survive).

In summary, **upregulated pathways** in the context of olaparib treatment fall into two categories: (1) those that **promote survival and DNA repair** (HR restoration, ATR/CHK1, PI3K, drug efflux), uniformly contributing to **drug resistance**, and (2) those that **promote anti-tumor immunity** (STING/IFN pathways), contributing to **drug sensitivity** especially in immune-reactive subtypes. Subtype-wise, TNBCs often exhibit both sides – they can activate immune pathways (good) but also adaptive repair/AKT pathways (bad) – whereas HR+/HER2– cancers primarily show the survival pathway upregulation (AKT), and HER2+ cancers are usually treated with other agents so data is limited.

Downregulated Pathways (by context/subtype)

Pathways found to be **transcriptionally or functionally downregulated** in the context of olaparib usually relate to the drug's action of disabling repair and proliferation. Key observations include:

- **Homologous Recombination Repair Downregulation:** The defining feature of olaparib-sensitive tumors is **loss or reduction of HR repair capacity**. In BRCA1-methylated triple-negative cancers, for example, the **BRCA1 gene is epigenetically downregulated**, causing an HRD phenotype that correlates with PARP inhibitor sensitivity ⁴⁴ ⁸ . Similarly, loss-of-function mutations in HR genes (BRCA2, PALB2, RAD51C, etc.) effectively **downregulate the HR pathway**, forcing reliance on alternative, error-prone repair. This is often **inherent in certain subtypes**: e.g. basal-like TNBC frequently has downregulated HR due to BRCA mutations or other HR defects, whereas luminal A cancers generally have intact HR. In patients, HRD is assessed by genomic “scars,” but functionally it equates to downregulation of HR function. Thus, **HR pathway downregulation** is a hallmark of responders (and is often pre-existing rather than caused by the drug). In trials, tumors with HRD (LOH-high, etc.) had better responses to olaparib than HR-proficient tumors ⁷ ⁴⁵ .
- **Cell Cycle Progression and Proliferation Pathways:** Responding tumors often exhibit **downregulation of proliferative gene sets** after treatment, reflecting cell-cycle arrest or cell death. For instance, genes in the **E2F targets** and **G2/M checkpoint** Hallmark sets may be suppressed in tumors on olaparib, since cells accumulate in S/G2 or undergo senescence/apoptosis. Clinically, a decrease in Ki-67 (a proliferation marker) was noted in some patients on olaparib, indicating reduced cell proliferation. This likely isn't a direct effect on cell-cycle regulators, but a consequence of DNA damage stress; nonetheless, one can consider **downregulation of cell-cycle/mitotic pathways** as part of olaparib's pharmacodynamics in sensitive tumors. In contrast, resistant tumors might maintain cell-cycle gene expression (or recover it quickly). Therefore, **loss of proliferation signaling** is generally associated with sensitivity (tumor growth slows or stops), whereas continued proliferation despite DNA damage signals resistance.
- **NHEJ and Alternative End-Joining:** In BRCA-mutant cells, the backup DSB repair via non-homologous end-joining (NHEJ) is normally active. However, **loss or downregulation of key NHEJ factors** (53BP1, RIF1, Lig4, etc.) can occur as a resistance mechanism – paradoxically **restoring HR**. From a pathway perspective, this means the **NHEJ pathway is functionally downregulated** in some resistant cancers (through mutations or low expression of NHEJ components), which allows more DNA end resection and usage of HR. For example, loss of 53BP1 (a negative regulator of end resection) has been documented in metastatic breast tumors that became PARP inhibitor-resistant ⁴¹ ²⁸ . So although NHEJ downregulation might sound like it impairs repair, in BRCA1-deficient cells it actually *frees up* resection for HR. Therefore, **downregulation of the 53BP1-RIF1-Shieldin NHEJ axis** is a pathway change that confers resistance. This is subtype-agnostic (driven by mutation selection under therapy), but most commonly observed in TNBC/BRCa tumors.
- **“BRCAness” Downstream Pathways:** Beyond DNA repair, HR-deficient tumors often have distinctive metabolic and transcriptional profiles. When those tumors are treated with olaparib, some of those pathways may further decline. For instance, **downregulation of MYC target pathways** has been seen in some contexts – possibly because cells can't sustain high metabolic demands under DNA damage stress. Another example: **downregulation of fork reversal proteins** like SMARCA1, which some resistant cells achieve to stabilize forks (resistance mechanism) ⁴⁶ ; that is effectively a

downregulation of a pathway (fork remodeling) leading to resistance. However, evidence of specific pathway downregulation is often indirect.

- **Estrogen Receptor Signaling:** In HR+ patients receiving olaparib (e.g. BRCA2-mutant ER+ cancers), concurrent endocrine therapy may be ongoing (as allowed in adjuvant setting ⁴⁷). Olaparib itself doesn't directly suppress ER signaling, but in some cases, **ER pathway activity may diminish** if the tumor is responding (since proliferation slows). Conversely, persistent ER-driven transcription could promote cell survival. No definitive data yet ties ER pathway downregulation to PARPi sensitivity, aside from the requirement that HR+ patients have had endocrine therapy (which aims to downregulate ER signaling). One could say that **effective downregulation of estrogen signaling (via hormone therapy)** is necessary to fully exploit olaparib in ER+ disease – as suggested by trial criteria ⁹. In practice, this means olaparib is used either after or alongside measures to downregulate ER (so indirectly, ER pathways are “down” when olaparib is active in HR+ cases).
- **Immune Suppressive Pathways:** Another angle – though olaparib can activate immune pathways, tumors that are initially immune-cold might have **high baseline expression of immunosuppressive cytokines or checkpoints**, which could get downregulated in responders. For instance, one study noted that **Myeloid-derived suppressor cell (MDSC) chemoattractants** like CXCL12 (SDF1) were reduced by olaparib, limiting MDSC recruitment ⁴⁸. Thus, **downregulation of immunosuppressive signals** (or of MDSCs in the TME) can occur and would favor sensitivity. This has subtype relevance: immune-cold luminal tumors have more to gain if those suppressive pathways are turned down, but the strongest effects are still seen in TNBC.

In summary, **downregulated pathways** in the context of olaparib generally relate to **loss of DNA repair or cell survival functions**: *on one hand, these can drive sensitivity* (HR down, cell-cycle down, immunosuppressors down), but *in certain cases, specific pathway downregulation can be a form of resistance strategy* (downregulating NHEJ to restore HR, or loss of pro-death signals like p53). In breast cancer subtypes: HRD-rich subtypes (TNBC) inherently have key pathways “down” (HR), making them sensitive, whereas an HR-proficient subtype would need artificial downregulation (via combination therapy) to be sensitized. The net effect depends on which pathway is downregulated and the biological context, as explored next in sensitivity vs resistance mechanisms.

Sensitivity Pathways (General Mechanisms)

Mechanisms of **sensitivity** to olaparib in breast cancer are largely rooted in the tumor's dependence on alternative DNA repair routes and certain genetic weaknesses. Broadly, breast cancers are **more sensitive** to olaparib when they harbor:

- **Homologous Recombination Loss:** Any gene loss-of-function that cripples HR repair will sensitize cells to PARP inhibition. Classic examples are **BRCA1 or BRCA2 mutations** (germline or somatic) ⁴⁹, which were the basis for olaparib's initial indication. Others include **PALB2 mutations, RAD51C/RAD51D mutations**, or **ATM loss** ⁶ ⁵⁰. Clinical data support this: e.g., a phase II trial showed **PALB2-mutated metastatic breast cancer** had high response rates to olaparib (similar to BRCA-mutant) ⁵¹. Likewise, tumors with BRCA1 promoter hypermethylation (no BRCA1 expression) often behave like BRCA-mutants and may respond. This category is pan-breast cancer but enriched in TNBC and certain HER2-negative subsets.

- **“BRCAness” or HRD-High Status:** Even without a single gene mutation, tumors with genomic signatures of HR deficiency (scarring) tend to be sensitive. For example, high HRD score (LOH-high) TNBCs respond better to olaparib ^{7 45}. These tumors often have downregulated HR pathways through combinations of alterations. **HRD+ status** is thus a predictive biomarker of sensitivity, recognized in guidelines for ovarian cancer and under study in breast cancer.
- **DNA Damage Response (DDR) Alterations:** Beyond HR, loss-of-function in other DDR pathways can increase reliance on PARP. **PTEN loss** was reported to cause HR defects (though controversial) and sensitize cells to PARP inhibitors in some studies ⁵². Similarly, deficiency in **FANCD2 or FANCA** (Fanconi anemia pathway) might confer sensitivity, as that pathway intersects with HR. These are less common in breast cancer but illustrate that any DDR weakness can be exploited by olaparib.
- **Replication Fork Instability:** Tumors unable to protect replication forks from collapsing are more sensitive to PARP inhibition. A key mediator here is **SLFN11** – if expressed, it prevents cells from pausing replication under stress, pushing them into lethal damage. While SLFN11 is more studied in SCLC and ovarian cancer, its presence in some breast cancers could enhance PARPi efficacy ²⁹. In essence, cancers lacking fork protection proteins (PTIP, EZH2, etc.) or with high replication stress will be hypersensitive to PARP inhibition ³⁰. TNBCs often have such features.
- **High Proliferation Rate:** Rapidly cycling cells accrue DNA damage faster and may be more dependent on PARP for repair of spontaneous lesions. Thus, a high Ki-67 (common in TNBC and some HER2+) might correlate with better responses, as the drug has more opportunities to induce lethal damage each S-phase. (This is a theoretical sensitivity factor; clinically, TNBCs with high proliferation do respond well to PARP inhibitors if HRD is present.)
- **Immune Activation (“hot” tumors):** An emerging concept is that **tumors with pre-existing T-cell infiltration or inflamed gene expression** might respond better to olaparib, especially in combination with immunotherapy. BRCA-mutant breast cancers often have higher TILs (BRCA1 TNBCs can be immune-“hot”), which could aid in clearing damaged cells. Additionally, olaparib can upregulate PD-L1 on tumor cells, so tumors that are already infiltrated by T-cells could benefit from that (though they might then need checkpoint blockade). While not a standalone sensitivity mechanism for monotherapy, an “immune hot” microenvironment in TNBC is thought to be synergistic with PARP inhibitors ^{24 25}.
- **Subtype Context:** Of the molecular subtypes, **basal-like and BRCA-like tumors (often TNBC)** are most sensitive, for the reasons above. **Luminal A** tumors are typically insensitive (unless a BRCA2 mutation is present) because they have intact HR and lower proliferation. **Luminal B** (more proliferative, sometimes HRD) could have intermediate sensitivity. **HER2-enriched** are not routinely treated with PARPi, but if a HER2+ tumor has HRD, it might be sensitive (this scenario is rare). Importantly, **triple-negative, BRCA1-mutant** breast cancers are exquisitely sensitive – this group drove olaparib’s FDA approval ¹. In gBRCA1/2 carriers, olaparib outperformed chemo in TNBC and ER+ patients alike ⁵³, but the absolute benefit was larger in TNBC (where no alternative targeted therapy exists and HRD is common).

To summarize general sensitivity mechanisms: any **upregulation of DNA damage (or inability to repair it)** and **lack of backup repair/ survival pathways** will render breast cancer cells highly sensitive to olaparib. The prototypical sensitive scenario is a **BRCA1-mutant TNBC** with HRD and a brisk immune infiltrate – such

a tumor is often completely reliant on PARP-mediated repair and will respond dramatically to PARP inhibition, as confirmed by clinical trials ³. On the other hand, an **ER+/HER2- tumor without HRD** is inherently insensitive (as seen in trials where non-BRCA patients had minimal benefit). The next sections detail specific pathways whose dysregulation (up or down) modulates sensitivity, with subtype annotations.

Sensitivity Pathways for Upregulation

Certain pathways, when **upregulated**, have been linked to **increased sensitivity** to olaparib. These often involve processes that promote cell death or augment the drug's effects. Key sensitivity-associated upregulations include:

- **cGAS-STING & Interferon Signaling – Upregulation of Innate Immune Response:** Elevated activity of the cGAS-STING pathway and downstream type I interferon signaling can enhance olaparib's anti-tumor effect. *Context:* Primarily in **BRCA-mutant TNBC**. *Rationale:* Olaparib-induced DNA breaks can activate cytosolic DNA sensors (cGAS/STING), leading to robust production of interferon-beta and chemokines ⁵⁴ ³⁸. This brings **CD8+ T cells** and NK cells into the tumor and heightens anti-tumor immunity. Upregulation of interferon-stimulated gene sets and antigen presentation pathways has been observed when STING agonists are combined with olaparib, yielding superior tumor control ³⁷ ³⁸. Even with olaparib alone, **gradual T-cell recruitment** was noted in BRCA1-deficient TNBC models (e.g. increased intratumoral CD8+ T cells by day 7) ⁵⁵. Clinically, this suggests that tumors with a pre-activated STING/IFN pathway or those able to mount an interferon response (often those with existing immune infiltration) respond better to PARP inhibition. In essence, **upregulation of immune inflammation** (a “hot” microenvironment) sensitizes tumors to olaparib by enabling immune-mediated clearance of damaged cancer cells ²⁴ ³⁶. This is subtype-specific: TNBC and some HER2+ tumors (with high TILs) could exploit this, whereas immunologically cold luminal tumors gain less benefit. Combination therapies (olaparib + PD-L1 or STING agonists) rely on this pathway's upregulation for maximal efficacy.
- **Pro-apoptotic Pathways – Upregulation of Apoptosis/Cell Death Signals:** When pathways that drive apoptosis are upregulated, olaparib sensitivity increases because cells more readily undergo programmed death after incurring DNA damage. *Context:* This can apply across subtypes, but may be more relevant in **ER+ tumors with functional p53**. *Rationale:* PARP inhibition alone doesn't induce apoptosis unless DNA damage is overwhelming or pro-death signals tip the balance. Upregulation of the **p53 pathway** (e.g. via restoration of wild-type p53 function, or via drugs that upregulate p53 targets) can enhance olaparib's effect – p53 will trigger cell-cycle arrest and apoptosis in response to DNA breaks ⁶. Similarly, higher expression of pro-apoptotic proteins like BAX, PUMA, or BIM can make cells more prone to die once PARP1 is inhibited. For example, **HDAC inhibitors upregulate pro-apoptotic genes** and have been shown to synergize with olaparib in TNBC by lowering the threshold for apoptosis ⁵⁶ ⁵⁷. Thus, any pathway that **increases apoptotic priming** (upregulating death receptors, caspases, etc.) will sensitize cells to the lethal DNA damage inflicted by PARP inhibition. Clinically, this concept underpins combinations like olaparib + chemotherapy (chemo upregulates intrinsic apoptosis) and is not subtype-restricted, though p53-mutant tumors rely on alternative cell death routes. In summary, **upregulation of apoptosis execution mechanisms** means olaparib-treated cells are less likely to arrest and repair, and more likely to irreversibly commit to death.

- **BRCA/HRD Signature Itself (Paradoxical Upregulation):** While loss of HR is the primary sensitizer, an interesting note is that some BRCA1-mutant cancers show **compensatory upregulation of alternative repair genes** that still don't fully rescue HR. For instance, BRCA1-deficient cells might upregulate PARP itself or other BER factors (a futile response, since PARP is inhibited by olaparib). This “BRCAness signature” involves upregulated DNA damage sensing (ATR, p53) but downregulated repair. It can be detected as a gene expression profile predictive of PARPi response ⁵⁸. Essentially, **upregulation of damage response (but not repair)** signals – like an activated p53 or ATM pathway in the absence of effective repair – correlates with sensitivity. Tumors appear “stressed” by accumulating DNA lesions and mount a partial response (checkpoint activation, etc.), which we can interpret as sensitivity markers. One could include **upregulated ATM->CHK2 signaling** leading to cell cycle arrest; if ATM is present but BRCA is not, ATM activation just halts cells, which may then die. This scenario might occur in some luminal BRCA2-mutated cancers where ATM is intact: olaparib causes ATM activation (DSB sensed) but no HR repair follows, leading to apoptosis – here ATM pathway up is a marker of effective damage induction.
- **Downstream Synthetic Lethal Partners – Upregulation of Mutant Pathway's Effect:** A concept in synthetic lethality is that sometimes an *upstream* pathway can be exploited. For example, **upregulation of the Myc oncogenic pathway** causes high replication stress; such cells depend heavily on PARP to repair frequent DNA lesions and are thus very sensitive to PARP inhibitors (Myc-overexpressing tumors have been proposed to be susceptible to PARP inhibition even without BRCA mutations) ⁵⁹ ⁴⁵. Though not clinically proven, if a breast tumor has *MYC* amplification (common in basal-like cancers) and thus high replicative stress, that upregulated oncogenic signal ironically creates a vulnerability to PARP inhibition. Thus, in a broad sense, **upregulation of any process causing DNA replication stress or damage** (c-Myc activation, Cyclin E overexpression, etc.) can sensitize cells to olaparib by increasing their reliance on PARP-mediated repair. This might help explain occasional responses in HR-proficient tumors that have other drivers of genomic instability.

Subtype nuances: TNBC often inherently exhibits many of these sensitivity-favoring features (HRD, Myc-driven stress, immune activation). HER2+ tumors, if they carry BRCA mutations, might also show strong responses (e.g., there are BRCA2-mutated HER2+ cases that responded in trials), potentially due to high proliferation plus HRD. In HR+ cancers, sensitivity usually requires deliberate induction (e.g., co-treat with PI3K inhibitor to create HRD, or high tumor grade with underlying genomic instability).

In summary, pathways whose **upregulation** boosts olaparib sensitivity are those that either **amplify the damage being done** (immune clearance, replication stress) or **promote the demise of the damaged cell** (apoptosis). Upregulating these processes tilts the balance against the cancer cell once PARP is inhibited.

Sensitivity Pathways for Downregulation

Several pathways, when **downregulated or suppressed**, lead to increased sensitivity to olaparib. In most cases, these are pathways that normally help the cell cope with DNA damage – their removal exposes lethal weaknesses. Key examples:

- **Homologous Recombination Repair – Downregulation of HR:** This is the quintessential sensitivity pathway. *Context:* All subtypes, demonstrated in **BRCA1/2-mutant and HRD** breast cancers. **Rationale:** Downregulating or losing HR function (through gene mutations, deletions, or epigenetic silencing) renders cancer cells unable to repair PARP inhibitor-induced double-strand breaks ⁶ ⁸.

This causes synthetic lethality. For example, **BRCA1 promoter hypermethylation** (transcriptional downregulation) in a TNBC will make that tumor exquisitely sensitive to olaparib ⁴⁴ ⁷. Similarly, **knockdown of RAD51 or BRCA2** in cell lines dramatically increases olaparib cytotoxicity. Clinically, any breast tumor with **downregulated HR gene expression** (via mutation or other mechanism) is a prime candidate for PARP inhibition. Thus, **loss or suppression of the HR pathway** is the most important downregulation leading to sensitivity. This is of particular relevance in TNBC and some high-grade ER+ tumors; it is required in HER2+ for any potential benefit.

- **Fanconi Anemia Pathway:** The FA pathway feeds into HR repair. Downregulation of FA genes (FANCD2, FANCA, etc.) also induces HRD and sensitizes to PARPis. In breast cancer, germline **FANCC** or **FANCD1 (BRCA2)** mutations exist in subsets; functional loss of these mimics BRCA loss. While rare, **downregulation of FA pathway** (which could be due to promoter methylation of FANCF, for example) has been associated with responsiveness to DNA-damaging agents including PARPis. Essentially, this is an extension of HR downregulation.
- **ATR/Checkpoint Signaling – Downregulation of ATR/CHK1:** In normal cells, ATR and CHK1 orchestrate the response to stalled forks, allowing repair or arrest. If this pathway is suppressed, cells cannot efficiently pause and repair; instead, they progress into mitosis with broken DNA, leading to mitotic catastrophe. *Context:* Preclinical combo studies, e.g. **ATR inhibitor + olaparib** in TNBC. **Rationale: Downregulation or inhibition of ATR** greatly magnifies PARP inhibitor lethality ⁶⁰ ⁶¹. Evidence: In TNBC models, ATR or WEE1 inhibition overcame PARP resistance – essentially by removing the cell's ability to cope with replication stress, forcing lethal outcomes ⁶¹. Therefore, **loss of ATR/CHK1 checkpoint control** is a sensitizing condition. Some tumors inherently have low ATR or mutations in ATM/CHK2; notably, **ATM-null tumors** (ATM being upstream of ATR in damage signaling) respond well to olaparib in prostate and likely breast ⁵⁰. In breast cancer, ATM is occasionally lost (around 2–3% of cases) – those might behave like HRD and indeed are included in some PARPi trials ⁵⁰. In summary, removing the “checkpoint/brake” on the cell cycle makes PARP-induced damage more deadly, so **downregulation of ATR/CHK1-mediated checkpoint** pathways increases sensitivity. This concept is being harnessed in trials combining olaparib with ATR, WEE1, or DNA-PK inhibitors in TNBC.
- **DNA Damage Tolerance Factors – Downregulation of Fork Protection:** Some proteins help cancer cells tolerate DNA lesions by protecting replication forks (e.g. **RIF1, PTIP, EZH2** recruitments). **Downregulating these proteins** can make cells more sensitive to PARP inhibitors because forks collapse more readily. A concrete example: **loss of PARP inhibitor resistance protein PARI (ADPRHL2)** or **downregulation of RADX** (a fork regulator) increases sensitivity by allowing uncontrolled fork degradation. Another is **PTIP**: its depletion was shown to independently increase cisplatin/PARP sensitivity by unbridled MRE11 activity at forks ⁶². While these are intricate molecular details, the principle is that **downregulation of fork-stabilizing pathways** (via gene loss or knockout) is sensitizing. In patients, such alterations might not be intentionally present before treatment, but could be induced by combination therapy (e.g. using an EZH2 inhibitor to downregulate fork protection and sensitize to olaparib – a strategy explored in trials ⁶³ ⁶⁴).
- **Oncogenic Pathways – Downregulation of Survival Signals:** Reducing pro-survival signaling can sensitize tumors. For instance, **downregulation of PI3K/AKT signaling** (through pharmacologic inhibitors or tumor suppressor activation) yields greater PARP inhibitor effect ³¹. In a trial, adding the AKT inhibitor capivasertib to olaparib showed activity even in non-BRCA mutant breast cancers

⁶⁵ . Mechanistically, inhibiting PI3K/AKT downregulates BRCA1/2 expression and induces a state of “BRCAness” ³¹ . So effectively, **downregulating the PI3K pathway** is a method of functionally downregulating HR (as mentioned earlier). Similarly, **downregulation of CDK4/6-Rb-E2F signaling** (via CDK4/6 inhibitors in ER+ cancers) could, in theory, synchronize cells in a phase where olaparib is more effective or reduce HR repair efficiency; some preclinical evidence suggests CDK inhibition may interfere with HR and sensitize to PARP inhibitors. Therefore, **turning off key survival and cell-cycle progression pathways** is often synergistic with olaparib. Clinically, this means that concurrent targeting of pathways like PI3K, EGFR, or anti-apoptotic proteins (BCL-2) – all forms of pathway “downregulation” – can heighten olaparib’s impact.

- **Tumor Microenvironment – Downregulation of Immunosuppression:** As mentioned earlier, reducing immunosuppressive factors (Tregs, MDSCs) can help. If a therapy (say, a CXCR4 inhibitor) **downregulates MDSC recruitment**, the immune-mediated component of olaparib’s effect may be stronger ⁴⁸ . Also, **downregulation of PD-L1 expression** (though counterintuitive, since olaparib tends to upregulate PD-L1) might allow T cells to function better without needing checkpoint inhibitors. In any event, a TME with fewer suppressive signals (which could result from interventions or tumor evolution) would pair well with PARP inhibitor therapy.

In summary, **pathways that, when suppressed, increase olaparib sensitivity** largely boil down to **removing DNA repair and cell survival options from the tumor**. The prime example is **HR pathway downregulation**, but also shutting off checkpoint and survival pathways like ATR and AKT. Many combination therapies in development for breast cancer (olaparib + PI3K inhibitor, olaparib + ATR inhibitor, olaparib + immune checkpoint blocker) are essentially attempts to **pharmacologically downregulate** those pathways and thereby **sensitize tumors** to olaparib. Subtype-wise, TNBC often already has some of these pathways naturally low (HR, etc.), which is why it’s most sensitive. HR+ tumors require deliberate downregulation (like adding a PI3K inhibitor to induce HRD, as being tried in trials).

Resistance Pathways (General Mechanisms)

Resistance to olaparib, whether inherent or acquired, arises through various mechanisms whereby cancer cells evade the lethal synthetic lethal interaction. Broadly, breast cancers exhibit **resistance** to olaparib via:

- **Restoration of Homologous Recombination:** The most frequent mechanism in BRCA-mutated cancers is to regain some HR repair capability. Tumors can acquire secondary **BRCA2 or BRCA1 reversions** (true back-mutations that restore the reading frame) ⁶⁶ ⁶⁷ . These mutations restore BRCA protein function, thus repairing DNA breaks that olaparib would otherwise exploit, leading to treatment failure. In an Annals of Oncology 2024 study, about 15–20% of patients who progressed on PARP inhibitors had acquired BRCA reversion mutations ⁶⁸ . Even without true reversions, cells can upregulate compensatory HR proteins or use alternative HR pathways (e.g. RAD52, POLθ for alternative end joining). The net effect is **HR pathway reactivation**, nullifying the synthetic lethality. This mechanism is seen across subtypes but obviously requires an HR-defective tumor to start with (so primarily TNBC or gBRCA carriers). If a luminal BRCA2 mutant tumor becomes resistant, BRCA2 reversion is a likely culprit.
- **Loss of 53BP1/Shieldin (NHEJ factors):** As described, BRCA1-deficient cells can restore HR by losing components of the 53BP1-RIF1-Shieldin complex. This **pathway of resistance** has been demonstrated in cell lines and patient samples ⁴¹ ²⁸ . Essentially, by **downregulating or mutating**

NHEJ-enforcing genes (TP53BP1, RIF1, REV7, or SHLD1/2/3), the cells allow extensive DNA end resection and RAD51 loading even without BRCA1, partially rescuing HR. This leads to PARP inhibitor resistance without fixing the original BRCA1 mutation. Clinically, analysis of metastatic biopsies post-PARP therapy found cases with acquired 53BP1 or RIF1 mutations ⁶⁹ ⁶⁷. This mechanism appears predominantly in BRCA1-mutant TNBC, consistent with BRCA1's role in end resection. It underscores the principle that **losing certain DNA repair “backup” pathways (NHEJ)** can paradoxically confer resistance by reinstating HR.

- **Increased Drug Efflux (MDR):** Some cancers become resistant via pharmacokinetic means – by pumping the drug out. Overexpression of **ABCB1 (P-glycoprotein)** is a known mechanism in cell line models of olaparib resistance ³⁹. This was first observed in a BRCA2-mutant mouse model: tumors that became resistant to olaparib had massively upregulated Abcb1a/b and could be resensitized by P-gp inhibitors ³⁹. In human ovarian cancer, recurrent ABCB1 fusions were found in ~16% of patients progressing on PARPi ⁷⁰, leading to overexpression. For breast cancer, evidence is more limited but similar mechanisms likely occur (especially as many patients receive taxanes, which select for P-gp overexpression, potentially cross-conferring olaparib resistance ⁴³). Thus, an **ABC transporter-mediated multidrug resistance pathway** is a plausible cause of primary or acquired resistance in breast tumors, particularly in chemotherapy-pretreated cases (common in TNBC). Importantly, this mechanism is agnostic to HR status – even an HRD tumor might not respond if insufficient drug reaches the nucleus.
- **PARP1 Mutation or Loss:** Another resistance mechanism is alteration of the drug's target. Rare **PARP1 mutations** in the DNA-binding or catalytic domain (e.g. a PARP1 E988K mutation) can render the enzyme less susceptible to being trapped by olaparib ⁴⁵. Alternatively, **loss of PARP1 expression** can also cause resistance – counterintuitively, if the cell doesn't rely on PARP1 at all, PARP inhibitors have less effect. Some resistant clones have truncated PARP1 that binds DNA but doesn't function (so no trap, no toxicity). While not widely reported in breast cancer patients yet, such mechanisms have been seen in vitro. In essence, if the **target pathway (BER/PARP1)** is removed or altered, the drug is less effective. However, completely losing PARP1 function has fitness costs, so this is a less common route than regaining HR.
- **Replication Fork Stabilization:** As noted, **enhanced replication fork protection** allows BRCA-mutant cells to survive without repairing breaks. Mechanisms include upregulation of **DNA polymerase theta (POLQ)** to anneal broken ends, stabilization of stalled forks by **ATR activation**, loss of fork degradation factors (e.g. **deletion of PTIP, inhibiting MRE11 activity**), or **RADX upregulation** controlling RAD51 at forks ²⁷ ³⁰. These changes mean that even though DSBs occur, the replication forks don't collapse as readily, giving cells more time to deal with damage. This is an active area of research. Clinically, it's hard to measure “fork protection” directly, but it correlates with poorer responses. For instance, an analysis found **EZH2 loss** (which normally helps recruit fork nucleases) in resistant tumors, pointing to fork stabilization as a driver of resistance ⁴⁶ ⁶². Interventions targeting this (like ATR or WEE1 inhibitors) are aimed at breaking this resistance.
- **Alterations in DNA Damage Signaling:** Some tumors might attenuate the initial damage signaling to avoid cell death. For example, if a tumor acquires a mutation that inactivates **p53** (if it wasn't already mutant), that can confer resistance by preventing apoptosis in response to PARP-induced DNA damage. Many BRCA-mutant TNBCs already have p53 loss, which might explain why they sometimes do not completely respond (the cells can undergo genomic chaos instead of orderly

apoptosis, potentially leading to resistance if they find a way to tolerate the chaos). Similarly, loss of **SLFN11** (which enforces replication arrest) was mentioned: SLFN11-low tumors can continue replicating under PARP inhibition and thus eventually develop resistance ²⁹. Essentially, if a cancer cell turns off pathways that would normally execute or signal fatal damage (like p53 or SLFN11 or TGF- β regulated cell death), it can survive longer and potentially evolve additional resistance.

- **MET/RTK Pathways & EMT:** There is some speculation that aggressive tumors might escape by changing phenotype. **EMT upregulation** could theoretically make cells more quiescent and drug-tolerant. Also, hyperactivation of receptor tyrosine kinases (RTKs) like MET could drive proliferation independent of DNA damage (and MET was shown in a prostate model to confer PARPi resistance via PI3K and ATR pathway activation ⁷¹). While data in breast are sparse, highly plastic tumors might survive PARP inhibition by entering a drug-tolerant persister state (often associated with EMT and stemness pathways up).

To summarize general resistance mechanisms: the tumor either **fixes the specific defect** olaparib is exploiting (restoring HR), **reduces drug impact** (pump it out or alter target), or **finds a workaround to avoid death** (fork protection, no apoptosis). In breast cancer patients, the dominant acquired resistance events observed are **BRCA reversion mutations** and possibly **53BP1 pathway loss** ⁶⁹ ²⁸. Primary (innate) resistance is seen in **HR-proficient tumors**, which simply lack the initial vulnerability (this is why biomarkers are crucial – an HR-proficient luminal cancer is intrinsically resistant, so we wouldn't give olaparib in that case). Below, we detail specific pathways whose upregulation or downregulation drive resistance, with evidence and contexts.

Resistance Pathways for Upregulation

Pathways that, when **upregulated**, lead to **olaparib resistance** typically allow the cancer cell to bypass the drug's effect or compensate for it. Important upregulated resistance pathways include:

- **Homologous Recombination Restoration – Upregulation of HR Activity:** Any **upregulation of HR repair capacity** will confer resistance. *Context:* Observed in **BRCA-mutated breast cancers** gaining resistance. **Rationale:** Upregulation can occur via **BRCA re-expression** (e.g. through a secondary mutation that restores BRCA1/2) or via increased expression of HR proteins like RAD51. In patient tumors, **co-occurring BRCA1/2 reversion mutations** have been documented at progression on PARP inhibitors ⁶⁸, functionally indicating HR pathway reactivation. Additionally, even without true reversion, resistant cells often show higher RAD51 foci formation (a marker of HR upregulation). For example, a resistant TNBC cell line might **upregulate RAD51** through gene amplification or demethylation, enabling DSB repair despite olaparib. **Subtype:** This pertains mainly to HRD tumors (TNBC, gBRCA carriers). In HR-proficient tumors, HR is already up and they're innately resistant; here we refer to previously HR-deficient tumors regaining HR. **Evidence:** Loss of 53BP1 (discussed below) is one route to HR upregulation; another is BRCA2 reversion seen in metastatic biopsy ⁶⁷. Therefore, **upregulation of the homologous recombination pathway** – by whatever molecular means – is a key upregulated resistance pathway.
- **53BP1-RIF1-Shieldin Pathway Down-regulation (Leading to HR Up):** (Included here because it results in HR upregulation.) As explained, **loss of 53BP1 or Shieldin** unleashes HR. So while NHEJ itself is down, the *effect* is an **upregulation of HR repair** that causes resistance ⁴¹ ²⁸. One could phrase it as “**negative regulators of HR are silenced**”, effectively **upregulating HR**. This

mechanism, seen in BRCA1-deficient settings, means that an NHEJ pathway *down* leads to an HR pathway *up* (so conceptually, HR upregulation is the resistance factor). It's worth noting in this section that **increased end resection and HR** due to 53BP1 loss is a common resistance finding. Subtype: mostly BRCA1-mutant TNBC.

- **PI3K/AKT/mTOR Pathway – Upregulation of Survival Signaling: Hyperactivation of the PI3K-AKT pathway** has been implicated in PARP inhibitor resistance in breast and ovarian cancer ³² ³¹. *Context:* This can be inherent in some HR+ cancers (many have PI3K mutations) or acquired (tumors can upregulate AKT as a stress response). **Rationale:** Active AKT signaling promotes cell survival under genotoxic stress and can also enhance HR repair through downstream effectors. One review noted that **PARP inhibition triggers AKT activation** as a cytoprotective feedback ⁷². In cell models, inhibiting AKT resensitized resistant cells, indicating that high AKT was enabling their survival. Moreover, AKT can phosphorylate and inactivate pro-apoptotic factors, meaning even if DNA damage occurs, cells with upregulated AKT avoid apoptosis. **Evidence:** Clinical studies have shown **synergy in combining PARPis with AKT inhibitors**, underscoring that AKT upregulation was a resistance factor being countered ⁶⁵. In breast cancer, **PTEN loss or PIK3CA mutation** (both upregulate PI3K/AKT) might correlate with weaker PARPi responses (this is being investigated). Therefore, **upregulation of the PI3K/AKT pathway** – often via **PTEN inactivation or PI3K mutation** – is a pathway leading to PARPi resistance, particularly relevant in **luminal cancers** (high PIK3CA-mutant frequency) and some TNBCs with PTEN loss.
- **PARP Hyperactivation & Drug Metabolism Pathways:** Some resistant cells show a paradoxical **upregulation of PARP itself and/or drug metabolism**. For instance, a resistant clone might massively express PARP1 (trying to overcome inhibition by sheer quantity) – though olaparib binding is competitive, beyond a certain point, some PARP activity might persist or the drug gets titrated out. Also, upregulation of enzymes that metabolize or sequester olaparib could occur (though olaparib's metabolism is mainly via liver CYP3A, not tumor). A related pathway is **upregulation of drug efflux pumps** (discussed separately below) – which is essentially upregulating a metabolism/elimination pathway in the cell.
- **Drug Efflux Transporters – Upregulation of ABCB1/ABCG2:** Perhaps the clearest non-HR mechanism: **overexpression of ABC transporters** leading to drug efflux ³⁹ ⁴⁰. *Context:* Seen in lab models; clinically suspected in heavily pretreated patients. **Rationale:** High levels of P-glycoprotein or BCRP on cancer cell membranes actively pump out olaparib, reducing intracellular drug concentration below effective levels. This upregulated pathway can cause cross-resistance to multiple drugs. **Evidence:** Nature studies in ovarian cancer cells showed **ABCB1 induction confers resistance to both paclitaxel and olaparib**, reversible by ABCB1 inhibition ⁷³. In breast cancer, an analysis of circulating tumor DNA after progression on olaparib found ABCB1 fusions (leading to overexpression) in some patients ³⁹. Subtypes: This is not specific to a subtype but may be more relevant in TNBC (which often requires chemo – selecting for MDR – prior to PARPi) and less so in untreated ER+ cases. It's also a concern when PARP inhibitors are considered in combination with other substrates of P-gp. In summary, **upregulation of the multidrug transporter pathway** (like the **REACTOME_ABCB1 transport**) is a well-established route to olaparib resistance.
- **Stromal and Growth Factor Signaling:** Tumor microenvironment can influence resistance. For example, **upregulation of IGF-1R or other growth factor signals** from stroma could promote HR repair or survival signaling in tumor cells, mitigating olaparib's effect. There's some evidence that

FGF and IGF signaling pathways get activated in PARPi-resistant ovarian cancer cells, conferring a growth advantage. In breast cancer, this hasn't been deeply explored, but one could suspect that **upregulation of alternative survival pathways (MAPK, EGFR, etc.)** might help bypass dependence on PARP. If a TNBC strongly upregulates EGFR pathway, that could drive proliferation despite DNA damage (EGFR can activate ATR via RAS->AKT too). These are speculative but plausible patterns of upregulation contributing to a drug-tolerant state.

- **Epithelial-Mesenchymal Transition (EMT) and Cell State Changes:** Some reports suggest that cancer cells that undergo EMT or acquire a stem-like state become less responsive to DNA-damaging agents. **Upregulation of EMT transcription factors (ZEB1, TWIST1)** could indirectly confer PARPi resistance by slowing proliferation (cells in a quasi-quiescent mesenchymal state) and activating stress survival programs. Also, EMT is associated with increased ABC transporter expression (linking to the efflux mechanism). While not specifically proven for olaparib, **EMT-upregulated pathways** (TGF- β , etc.) generally correlate with therapy resistance and could be relevant. Subtype: TNBC, especially claudin-low subtype, inherently has more EMT traits and might show lower PARPi sensitivity unless HRD is present.

In essence, **resistance-upregulating pathways** allow cancer cells to **repair more, survive better, or avoid the drug**. Upregulation of HR repair and survival (AKT) are central in BRCA-mutant contexts, whereas upregulation of drug efflux and generic survival signals can affect any cancer. Identifying these in patients is challenging; however, translational studies have pinpointed HR restoration and ABCB1 as frequent culprits in those who relapse on olaparib ⁶⁸ ³⁹ .

Resistance Pathways for Downregulation

Pathways whose **downregulation** leads to **resistance** are those that, when lost, remove a vulnerability that olaparib was exploiting or activate a compensatory mechanism. Key pathways include:

- **Non-homologous End-Joining (NHEJ) – Downregulation of 53BP1/Shieldin:** As discussed, **loss (downregulation) of 53BP1 or Shieldin complex** subunits in BRCA1-deficient cells reinstates HR, causing resistance ⁴¹ ²⁸ . *Context:* Acquired resistance in **BRCA1 mutant TNBC**. **Rationale:** The 53BP1-RIF1-Shieldin pathway normally blocks DNA end resection; its downregulation means ends are resected and RAD51 can load even without BRCA1, partially restoring HR repair of DSBs ²⁸ . This essentially negates olaparib's synthetic lethality by allowing an alternate repair route. **Evidence:** In patient tumors, low 53BP1 expression has been found after PARP inhibitor therapy, and in vitro deletion of REV7 or Shieldin genes causes PARPi resistance ²⁸ ⁷⁴ . Thus, **downregulating the NHEJ pathway** that was restraining HR is a clear resistance mechanism. This is a specific case where a *downregulation* (of NHEJ) leads to an *upregulation* of HR (resistance).
- **Apoptosis Pathway – Loss of p53 or Pro-apoptotic Factors:** If a tumor cell downregulates its cell-death pathways, it can survive DNA damage that would normally kill it. *Context:* Many BRCA-associated cancers already lack p53 (which might actually promote genomic instability and initial sensitivity, but at the point of cell death, p53 helps execute death). **Rationale: TP53 mutation/downregulation** means cells do not undergo apoptosis or permanent arrest in response to DNA damage, potentially allowing them to continue dividing until they acquire HR-restoring changes. In one analysis of PARP-resistant ovarian cancer, all resistant tumors had dysfunctional p53 (though so did sensitive ones, so it's necessary but not sufficient). Similarly, downregulation of pro-apoptotic

proteins (BAX, etc.) or upregulation of anti-apoptotics (BCL-xL) can let cells repair or tolerate more damage. In summary, if the **apoptosis pathway is suppressed**, cells can evade olaparib's lethal effect, resulting in treatment resistance. *Subtype*: TP53 is mutated in ~80% of TNBC and ~20-30% of luminal cancers; those with wild-type p53 may respond initially but could acquire p53 dysfunction to resist further (less common than BRCA reversion though).

- **DNA Damage Sensing – Loss of SLFN11 and Other Mediators**: **SLFN11** is a protein that ties DNA damage to blocking replication and inducing cell death. Tumors with high SLFN11 are very sensitive to DNA-targeted therapies, whereas SLFN11-negative tumors can resist by continuing to proliferate slowly despite damage ²⁹. If a breast tumor that originally expressed SLFN11 loses it (downregulation via epigenetic silencing, for example), it may become more resistant to olaparib because it no longer forces replication shutdown upon PARP trapping. Similar logic applies to other sensors: e.g., if a cell downregulates ATM excessively (complete loss) it might reduce initial signaling of damage and rely entirely on backup microhomology end-joining, potentially making it resistant if the context allows cell survival (though ATM loss usually sensitizes; it's a nuanced case: ATM loss helps initial response but can drive an unstable phenotype that finds other escapes). On balance, **losing positive mediators of cell death in response to damage (like SLFN11)** is a resistance move.
- **Downregulation of PARP Itself (Target Loss)**: Surprisingly, **loss of PARP1 expression** can cause resistance. This was shown in a few cases where BRCA-mutant cells spontaneously lost PARP1 – those cells were less affected by PARP inhibitors because there was less drug target, and they also tended to rely on alternative repair that doesn't need PARP1. In lab models, PARP1 knockout cells are not killed by olaparib (there's nothing to trap), though they have other DNA repair issues. In patients, complete PARP1 loss is rare (since tumors likely need some PARP function to survive in the first place). But a truncating mutation in PARP1 that prevents trapping can effectively be like a functional loss and confer resistance ⁴⁵. So, **downregulation or inactivation of the drug's target pathway (PARP1-BER)**, while counterintuitive, can underlie resistance.
- **Miscellaneous Metabolic Changes**: There's speculation that cancer cells might adapt their metabolism to mitigate PARP inhibitor effects – e.g., increasing NAD⁺ production to fuel remaining PARP activity, or shifting cell cycle distribution (some might go into G0/quiescence, effectively avoiding S-phase where olaparib is most toxic). If a subpopulation of cells downregulates pathways driving proliferation (semi-dormant persists), they can wait out the drug – a form of transient resistance. This overlaps with EMT/stem cell state changes as mentioned.

To highlight a concrete example of resistance via pathway downregulation: A recent study (“Longitudinal profiling... Ann Oncol 2024”) found that in metastatic breast cancer resistant to PARP inhibitor, aside from BRCA reversions, mutations in **PAXIP1** (PTIP) were present ⁶⁹ ⁷⁵. PTIP is involved in recruiting nucleases to stalled forks; losing PTIP **downregulates a fork degradation pathway**, resulting in more fork stability and resistance ⁶². This illustrates how loss of a factor (PTIP) that normally made PARP toxicity worse (by chipping away at forks) ends up making the cell more PARP-tolerant.

In summary, **resistance via downregulation** often means **removing a pro-death or pro-damage pathway**. Cancer cells shut off certain genes (53BP1, PTIP, p53, SLFN11, etc.) that were contributing to olaparib's efficacy, thereby escaping its lethal effects. These mechanisms are usually acquired under drug pressure and highlight the dynamic nature of resistance evolution.

Breast Cancer Subtype–Stratified Evidence

Different breast cancer subtypes exhibit distinct behaviors with olaparib, rooted in their molecular characteristics:

- **Triple-Negative Breast Cancer (TNBC):** TNBC is the subtype most associated with olaparib benefit, especially if BRCA1-mutated or HRD-positive. **Evidence:** In the OlympiAD trial, the majority of BRCA-mutated patients were TNBC, and they derived significant PFS benefit from olaparib ⁵³. TNBC often harbors BRCA1 mutations (germline or somatic) and exhibits “BRCAness” even without mutation (about 10–20% of sporadic TNBC have HRD) ⁷⁶. Furthermore, TNBC tends to have higher proliferation and sometimes higher immune infiltration, both potentially augmenting olaparib response (with or without combination therapy). Subgroup analyses suggest that **BRCA-mutated TNBC** responds better than **BRCA-mutated HR+** tumors in terms of absolute risk reduction, likely because TNBC has fewer alternative pathways (no ER signaling to rescue it) and a more pronounced HRD phenotype. TNBC also encompasses basal-like (BRCA1-like) and claudin-low (EMT) subtypes; the former responds strongly to PARP inhibition, while the latter might be less inherently HRD but could still benefit if HRD is present. Additionally, PD-L1-positive TNBC is being studied with olaparib + immunotherapy (e.g., MEDIOLA trial) with encouraging results, leveraging the immune pathway upregulation by olaparib ²⁴. In summary, TNBC patients with gBRCA1/2 or high HRD are prime candidates, and even TNBC without those biomarkers are subject of ongoing research for combination strategies. If a TNBC is BRCA-wild-type and HRD-low, it generally does not respond to olaparib monotherapy (as seen in negative parts of trials like GeparOLA which tested HRD+ vs HRD–TNBC).
- **HR+/HER2– (Luminal A/B) Breast Cancer:** By itself, the HR+ subtype (especially Luminal A) has low HRD frequency (~2-3%). Olaparib is **approved in HR+ gBRCA carriers** but only after endocrine therapy has been utilized ⁹. Evidence from OlympiAD: HR+ cohort (which was about ~50 patients) also benefited, but those were all gBRCA2 (mostly) or gBRCA1 mutant. They had to have progressed on hormonal therapy. The presence of ER signaling doesn’t directly impede olaparib, but these tumors often grow slower and have intact HR, making them less reliant on PARP. Luminal B tumors (high grade HR+) have a bit more genomic instability; indeed some gBRCA2 cancers present as ER+. These can do well with olaparib – for instance, in OlympiAD, ER+ patients with gBRCA had a PFS benefit comparable to TNBC group ⁵³, although cross-trial comparisons suggest that adding olaparib to endocrine therapy might not be broadly beneficial without a mutation. In early breast cancer, **adjuvant olaparib** is recommended for **HER2-negative, BRCA-mutant patients** regardless of ER status ⁷⁷, but with different criteria: ER+ patients had to have ≥ 4 nodes (i.e., high risk) ⁷⁸ whereas TNBC needed tumor >2 cm or any node ⁷⁸. This reflects that ER+ BRCA cancers generally have slightly better prognosis, so narrowing to very high-risk ones for adjuvant PARPi. Summarily, in HR+ breast cancer, **olaparib is effective if a germline BRCA mutation is present**, but has no role in biomarker-negative cases. There is ongoing research on combining PARP inhibitors with endocrine therapy or CDK4/6 inhibitors for HR+ disease, but nothing practice-changing yet. One note: HR+ BRCA2-mutant cancers have high rates of response to olaparib in metastatic setting, but often these patients would also be on endocrine therapy which can continue with olaparib (since mechanisms differ) ⁴⁷. Tolerability in combination is an area of interest, as is whether prolonged suppression of HR (via estrogen deprivation) could synergize.

- **HER2-Positive Breast Cancer:** These patients were excluded from major PARP trials unless they also had a BRCA mutation. BRCA mutations in HER2+ patients are relatively rare, but they do occur. The OlympiA adjuvant trial and FDA indication specifically say **HER2-negative**, so olaparib is not standard for HER2+ even if BRCA-mutated, because those patients weren't in the trials. However, case reports and subgroup analyses from other studies (like the TBCRC048 exploratory trial) suggest that **BRCA-mutant, HER2+ patients** can benefit from PARP inhibitors after progressing on HER2-targeted therapies ⁷⁹ ⁸⁰. The biology: if HER2 is driving the tumor strongly, one would target HER2 first. If that patient also has a BRCA mutation, PARPi could be used in later lines or concurrently in trials. One such trial is examining olaparib with HER2 blockade in HER2+ BRCA-mutant metastatic breast cancer. For HER2+ without BRCA, there's no evidence to use olaparib – HRD is uncommon there except in some basal-HER2 experiments. So in summary, **HER2+** is not an approved setting for olaparib *unless* there is a germline BRCA mutation; in that scenario it's a consideration after standard HER2 therapy. Real-world use is limited, and those tumors might have unique resistance interplay between HER2 signaling and DNA repair (some preclinical data hinted HER2 activation could upregulate RAD51, but targeting HER2 would take precedence clinically).
- **gBRCA1/2-Mutated / HRD-Positive:** This is a biomarker-defined “subset” that cuts across clinical subtypes. **Any breast cancer** (HER2-, mainly) with a germline BRCA1 or BRCA2 mutation falls in this category. The presence of the mutation predicts high sensitivity to olaparib ¹. OlympiAD and OlympiA trials are essentially analyses of this subgroup (with stratification by ER status). **In metastatic setting:** gBRCA-mutated patients had ~60% objective response to olaparib vs ~29% to chemo in OlympiAD ³. **In early setting:** adjuvant olaparib significantly improved 3-year invasive disease-free survival (85.9% vs 77% placebo) ⁸¹ ⁸². These benefits are irrespective of hormone receptor status, though more TNBC were BRCA1 and more ER+ were BRCA2. Somatic BRCA mutations are rarer but are likely similarly predictive; ongoing trials (e.g., Niraparib in somatic BRCA metastatic breast cancer) suggest activity, though olaparib's label doesn't yet include somatic BRCA in breast (it does in other cancers). **HRD-positive (non-BRCA):** In ovarian cancer, HRD testing can qualify patients for PARPi; in breast, this approach is not yet standard but being explored. The TBCRC048 phase II study found that some **non-BRCA HRD-high metastatic breast cancers responded to olaparib or talazoparib**, especially those with PALB2 mutations (ORR ~80%) ⁸³. However, those with solely high genomic instability without an identifiable mutation had more modest response (~30%). So HRD as a broader biomarker is promising but not fully validated in breast cancer; nonetheless, it's plausible that in coming years, **HRD-positive TNBC** (even without BRCA) might be considered for PARP inhibitors, particularly in combination with other therapies. For now, guidelines focus on gBRCA.
- **Basal-like vs Luminal vs HER2-enriched vs Claudin-low:** On the molecular level (PAM50 intrinsic subtypes), **Basal-like** correlates with TNBC and BRCA1-ness. Basal-like tumors have high prevalence of TP53 mutations and frequent DNA repair defects, making them prime candidates for PARPi. **Luminal A** (ER+, low grade) rarely have HRD and thus rarely benefit. **Luminal B** (ER+, high grade) occasionally have BRCA2 or other HRD features, so a small subset might. **HER2-enriched** (HER2+, non-luminal) can have high proliferation and sometimes co-mutations (if HRD, they'd behave like basal in response). **Claudin-low** (an EMT subtype of TNBC) often has low proliferation and immune infiltration; these might be less sensitive unless a BRCA mutation is present, due to intrinsic drug-tolerant features (and indeed, not all TNBC respond uniformly – some claudin-low TNBC cell lines are relatively resistant to PARP inhibitors despite BRCA1 mutation, perhaps requiring combination therapy).

- **Clinical Stage and Context:** In metastatic disease, PARP inhibitors are typically used as single agents in later lines (after chemo for TNBC, after endocrine for HR+). In that setting, the competition is chemo: OlympiAD showed olaparib was superior to more chemo in gBRCA patients ³. In early-stage (adjuvant), olaparib is given after standard chemo to mop up residual disease in gBRCA patients ². Notably, the benefit was seen in both TNBC and HR+ (with the risk criteria differences) ⁷⁸. Adherence might be easier in HR+ because they continue endocrine therapy concurrently, whereas TNBC patients just finish and then take a year of olaparib (with manageable toxicity). The different natural histories (TNBC relapses earlier, HR+ later) also mean the impact of olaparib may play out differently in long-term follow-up (we expect OS benefit in TNBC earlier; in HR+ it might emerge later).

In conclusion, **subtype matters enormously** in identifying who should get olaparib and how they respond: **BRCA-mutant basal-like/TNBC** is the archetype of responsiveness, **HR+ or HER2+ without HRD** essentially do not benefit and are not indicated, and **BRCA-mutant luminal** cancers do benefit but must be high-risk enough to justify PARPi (because they have other therapy options too). These nuances align with trial data and current clinical guidelines.

Contraindications and Safety

Olaparib is generally well-tolerated, but there are important safety considerations and formal contraindications drawn from regulatory labels:

- **Absolute Contraindications:** According to FDA and EMA labeling, there are **no absolute contraindications** to Lynparza aside from known severe hypersensitivity to olaparib or its excipients ⁸⁴ ⁸⁵. The US Prescribing Information lists “None” under contraindications ⁸⁶. However, **EMA** explicitly contraindicates **breastfeeding during treatment and for 1 month after the final dose** ⁸⁷, given the potential for serious adverse reactions in the nursing infant (this is effectively a contraindication to use in lactating women).
- **Pregnancy and Teratogenicity:** Olaparib can cause embryo-fetal harm (teratogenic and embryocidal in animal studies) ⁸⁸ ⁸⁹. It is Pregnancy Category D. While not phrased as “contraindicated” (since treating pregnant cancer patients is usually avoided anyway), in practice **use in pregnancy is contraindicated** due to the mechanism of action and animal data ⁸⁸ ⁸⁹. Women of childbearing potential should use effective contraception during treatment and for a period after (6 months after last dose per EMA). Men with pregnant partners or trying to father a child are also advised to use contraception (3 months after last dose for men) because olaparib may affect sperm and embryo viability.
- **Breastfeeding:** As noted, **contraindicated during therapy and 1 month after** ⁸⁷. It is unknown if olaparib is excreted in human milk, but due to potential toxicity (it’s a cytotoxic drug), nursing must be discontinued or the drug not started.
- **Hepatic and Renal Impairment:** **Severe hepatic impairment (Child-Pugh C)** – not studied; use is *not recommended* (cautionary avoidance) ⁹⁰. Mild to moderate hepatic impairment did not require dose adjustment ⁹⁰. **Severe renal impairment (CrCl <30 mL/min)** – not studied and thus not recommended ⁹¹. Moderate renal impairment (CrCl 31–50) requires dose reduction to 200 mg BID ⁹². These are not absolute contraindications, but dose adjustments or avoidance are advised. So in

practice, **patients with end-stage renal disease or Child-Pugh C liver disease should not receive olaparib** unless benefit outweighs risk and close monitoring is feasible.

- **Concomitant CYP3A4 Interactions:** Strong CYP3A4 inhibitors or inducers are to be avoided with olaparib. For example, co-administration of rifampin (a strong inducer) could significantly reduce olaparib exposure – while not a labeled “contraindication,” it is a **strong warning**. Similarly, grapefruit or Seville oranges (CYP3A inhibitors) can increase olaparib levels and should be avoided ⁹³. Patients on phenytoin, St. John’s Wort, etc., should switch off those before starting olaparib.
- **Myelodysplastic Syndrome / Acute Myeloid Leukemia:** Olaparib carries a warning for rare cases of therapy-related MDS/AML. In clinical studies, <1.5% of patients on long-term PARPis developed MDS/AML, often those heavily pretreated with chemo ⁹⁴ ⁹⁵. This is not a contraindication, but a caution: patients should be monitored (baseline and periodic blood counts) ⁹⁶. If prolonged cytopenias occur, drug should be interrupted and bone marrow evaluation done ⁹⁷. If MDS/AML is confirmed, olaparib must be permanently discontinued ⁹⁸. Essentially, **previous therapies** (like platinum and anthracyclines) plus PARP inhibition can predispose to MDS. While not contraindicated in those with baseline cytopenias, one should correct severe anemia or neutropenia before starting.
- **Other Warnings/Precautions:**
 - **Pneumonitis:** Rare (<1%) cases of pneumonitis, including fatalities, have been reported ⁹⁹. Any new or worsening respiratory symptoms require interruption and investigation; if drug-induced pneumonitis is confirmed, permanently discontinue ¹⁰⁰. Patients with risk factors for interstitial lung disease should be monitored closely.
 - **Venous Thromboembolism:** An increase in pulmonary embolism incidence was noted in metastatic castration-resistant prostate cancer patients on olaparib + abiraterone ¹⁰¹. In breast cancer patients, VTE risk wasn’t highlighted, but clinicians should be vigilant, especially in high-risk individuals.
 - **Gastrointestinal Toxicity:** Nausea, vomiting, and diarrhea are common (>20%). These are manageable with supportive meds and typically not dose-limiting in most patients ⁵. No specific contraindication, but patients who cannot keep oral medication due to intractable vomiting might not tolerate therapy.
 - **Bone Marrow Suppression:** As a cytotoxic, olaparib commonly causes anemia (up to ~40% any grade, ~18% ≥Grade 3 in OlympiAD ¹⁰²), leukopenia/neutropenia (~15–20%), thrombocytopenia (~5–10%). It’s not contraindicated in patients with mild cytopenias (they often have some from prior chemo), but it should be used cautiously in those with significant bone marrow impairment. Baseline CBC and monthly CBC monitoring for 12 months (per EMA) is recommended ⁹⁶. If patients have baseline Grade 4 cytopenia, one might defer treatment until improvement.
 - **Combination with other myelosuppressive agents:** Not a formal contraindication, but overlapping toxicity requires caution (e.g. if considering olaparib with chemo – generally not done outside trials due to enhanced toxicity).
- **Population-Specific:**
 - **Pediatric:** Not approved; safety unknown.

- **Geriatric:** No specific contraindication; clearance slightly lower in older patients but no dose adjustment purely based on age.
- **Prior therapies:** Patients who have progressed on a platinum chemotherapy – there's cross-resistance between platinum and PARPi in BRCA-mutant disease. It's not a contraindication, but responses are shorter if the tumor is platinum-refractory. Some guidelines suggest not using PARPi immediately after platinum if the tumor progressed on platinum (because likely the tumor has a HR restoration). This is more a clinical consideration than a contraindication.

In practice, **before starting olaparib**, clinicians ensure: the patient is not pregnant; not breastfeeding; has adequate blood counts, renal, and liver function (or adjustments made); and isn't on interacting meds. Patients are counseled on contraception and to avoid certain foods/drugs.

Safety profile highlights: Most common side effects ($\geq 10\%$) include **nausea, fatigue/asthenia, anemia, vomiting, diarrhea, dysgeusia, decreased appetite** ⁵. These are usually Grade 1–2 and manageable (antiemetics for nausea, transfusions or dose reduction for anemia). There is no black-box warning in the US, but the key warnings are for MDS/AML and pneumonitis as above ⁸⁶ ⁹⁴.

Contraindications per guidelines: NCCN and ASCO guidelines do not list additional contraindications beyond label – they emphasize patient selection by biomarker and prior therapy rather than comorbidities. For instance, NCCN notes to avoid olaparib in patients with active uncontrolled infection or immediately after major surgery, etc., but those are general chemo precautions.

In summary, **absolute contraindications:** hypersensitivity, breastfeeding, and effectively pregnancy. **Relative contraindications/cautions:** severe organ impairment (renal/hepatic), ongoing use of strong CYP3A modulators, and baseline severe cytopenias. Olaparib's safety is well-characterized; with proper patient selection and monitoring, it is a safe addition to the breast cancer armamentarium.

Trial and Guideline Context

Clinical Trials: The use of olaparib in breast cancer is supported by pivotal trials and many ongoing studies:

- **OlympiAD (NCT02000622):** Phase III for metastatic HER2-negative breast cancer with germline BRCA1/2 mutations ¹⁰³. Patients had received ≥ 1 prior chemo (and endocrine if HR+). **Result:** Olaparib monotherapy significantly improved PFS vs physician's choice single-agent chemo (median 7.0 vs 4.2 months, HR 0.58) ³. Objective response rate ~59.9% vs 28.8% ³. Benefit was seen in both TNBC and HR+ subsets, although PFS HR was slightly more favorable in TNBC (~0.43) than HR+ (~0.82) in subgroup analysis (not statistically powered) – consistent with TNBC having fewer alternatives and higher HRD. OlympiAD established olaparib as a new standard option in gBRCA-mutated MBC, leading to FDA approval in January 2018 ¹⁰⁴.
- **EMBRACA (Talazoparib trial):** Not about olaparib, but similarly showed PARP inhibitor benefit in gBRCA MBC. It reinforces the class effect and influenced guidelines to list "PARP inhibitor (olaparib or talazoparib)" as preferred options in that population.
- **OlympiA (NCT02032823):** Phase III for **adjuvant** therapy in high-risk early breast cancer with germline BRCA1/2 mutation ². Patients had TNBC ($\geq T2$ or $\geq N1$) or HR+/HER2- (≥ 4 nodes) and had completed chemo (neoadjuvant or adjuvant). Olaparib 1 year vs placebo post standard therapy.

Result: Statistically significant improvement in 3-year invasive disease-free survival (85.9% vs 77.1%) and distant DFS ¹⁰⁵, and an early overall survival benefit (HR 0.68) ¹⁰⁵. This trial led to FDA approval in March 2022 for adjuvant use ² and incorporation into guidelines as a category 1 recommendation for this population ¹⁰. This is notable as one of the first targeted therapies in early TNBC beyond chemo. Ongoing follow-up will clarify long-term OS benefit (which appears significant already).

• **Other Trials:**

- **MEDIOLA** (Phase I/II): Olaparib + Durvalumab in gBRCA metastatic breast cancer (especially TNBC cohort) ¹⁰⁶. Interim results showed encouraging response rates, suggesting synergy (ORR ~63% in small TNBC cohort). This combination leverages immune activation by olaparib and checkpoint blockade.
- **KEYLYNK-009** (Phase II/III): Pembrolizumab + Chemo induction, then randomize to Olaparib + Pembrolizumab vs Chemo + Pembrolizumab maintenance in metastatic TNBC (regardless of BRCA). This Merck trial (MK-7339 is olaparib) is exploring broader use in TNBC by combining with immunotherapy (no results yet public).
- **Olaparib + PI3K inhibitor (alpelisib)**: A phase I (PETRA) and others tested this in TNBC and ovarian. In a Lancet Oncol 2019 study, olaparib + alpelisib had some activity in platinum-resistant BRCA wild-type cancers ¹⁰⁷ ³³, hinting at synergy by inducing HRD via PI3K inhibition.
- **Olaparib + ATR inhibitor (ceralasertib)**: The CAPRI trial and an cohort in the plasmaMATCH trial (cohort E) examined this in advanced breast cancer, especially TNBC with prior PARP or platinum. Results (from plasmaMATCH cohort E) showed modest activity: disease control in some patients, but also significant toxicity (myelosuppression). It's proof-of-concept that ATR inhibition can overcome resistance in some cases.
- **Talazoparib + Enzalutamide** in AR-positive TNBC (ABRAZO) – exploring PARP in unusual subtypes.
- **Neoadjuvant trials**: A small phase II (GeparOLA) tested adding olaparib to chemo vs chemo alone in early HRD-positive HER2-negative breast cancer. It found higher pCR (56% vs 0% in HRD-high non-BRCA patients for chemo+olaparib vs chemo alone, albeit small numbers), suggesting a benefit in the neoadjuvant setting for HRD tumors. This hints that beyond germline BRCA, HRD selection might identify who benefits from PARP in earlier disease.
- **Basket trials**: Olaparib has been tried in other settings like metastatic breast cancer with somatic mutations in HR genes (Niraparib's TBCRC048 and an ongoing olaparib Expanded Indication trial). Preliminary data: PALB2-mutated breast cancers respond almost as well as BRCA (some complete responses), while others like ATM or CHEK2 had lower response rates ⁸³.

Guideline Recommendations:

- **NCCN (v4.2023 Breast Cancer Guidelines)** – For **metastatic HER2-negative, gBRCA1/2-mutated** breast cancer, NCCN lists olaparib (or talazoparib) as a **preferred treatment** in the first-line or subsequent lines, provided the patient has no prior PARP exposure ¹⁰⁸. It notes that if HR+ BRCA-mutated, patient should also have endocrine-resistant disease (per OlympiAD criteria) before using PARP inhibitor. NCCN also now includes a section on **adjuvant therapy for BRCA carriers**: “For patients with high-risk, HER2-negative early breast cancer and a germline BRCA mutation, consider one year of adjuvant olaparib (category 1)” ¹⁰⁹. This is a direct adoption of OlympiA results. The NCCN guidelines emphasize testing for germline BRCA in appropriate patients (all TNBC ≥T1 or any TNBC at diagnosis, and in high-risk HR+ under 45 or with family history, etc.) to identify who may benefit from PARP inhibitors ¹¹⁰.

- **ASCO Guidelines (2020, updated 2021)** – ASCO's rapid update (July 2021) recommends offering 1 year of adjuvant olaparib to patients with HER2-negative early breast cancer, gBRCA mutation, who meet OlympiA criteria ¹⁰ ⁷⁷. ASCO also recommends

germline genetic testing for anyone with HER2-negative breast cancer ≤ 60 years (which captures most TNBC) and any patient with family risk factors, to find BRCA mutations. For metastatic, ASCO guidelines (2019) already stated that a PARP inhibitor is preferable to chemo for patients with gBRCA metastatic breast cancer who have previously been treated with anthracycline and taxane (and endocrine if ER+) ^{79 111}. ASCO and ESMO both highlight that if a patient has platinum-resistant disease or progressed on platinum, PARP inhibitors might have diminished efficacy (cross-resistance), but can still be considered. - **ESMO (2019, 2020)** – ESMO's metastatic breast cancer guideline included olaparib as a recommended option for gBRCA-mutated, HER2-negative cases after prior therapy. ESMO Early Breast Cancer 2021 (post-OlympiA) incorporated adjuvant olaparib similarly for gBRCA, high-risk patients (level I evidence). - **NICE (UK)** – In 2019, olaparib was approved by NICE for gBRCA-mutated advanced breast cancer after ≥ 1 chemo (with a note that cost-effectiveness was acceptable using a managed access scheme). NICE guidelines reflect similar criteria: HER2-; prior therapy done; use olaparib in line of therapy per trial. In 2022, NICE also rapidly approved adjuvant olaparib for gBRCA, high-risk early breast cancer, making it standard in the UK as well. - **Other regions** – Health Canada and Australian TGA approved olaparib in these settings as well, aligning with FDA/EMA. The Japanese NCCN-equivalent (JSCO) also added olaparib for gBRCA metastatic and adjuvant after OlympiA (Japan participated in the trial).

Combination in Guidelines: Currently, guidelines do not yet recommend combination of PARP inhibitors with other agents outside of a trial (except in prostate cancer, where olaparib is combined with abiraterone – but that's a different disease). However, NCCN mentions clinical trial options such as PARP + PD-1 for TNBC, reflecting ongoing research.

Sequencing: For metastatic TNBC gBRCA, NCCN suggests a PARP inhibitor can be used as an alternative to platinum chemotherapy, as both have similar efficacy in that population ¹¹². Some clinicians prefer platinum first (cheaper, IV), others prefer PARP inhibitor (oral, better QoL). There's an ongoing question: in BRCA carriers, is platinum $\rightarrow$ PARP sequentially better or vice versa? Studies like BROCADE (veliparib with chemo) tried combo, but no definitive answer. Guidelines generally leave it to clinical judgement but ensure that a patient gets one or the other.

Real-world context: Olaparib has been in use for a few years, and data from expanded access and real-world cohorts confirm its efficacy and manageable toxicity in BRCA-mutated breast cancer, consistent with trial outcomes. Adherence to the full year in adjuvant setting in trials was fairly good (~70% completed 1 year), though dose reductions were needed in ~25% due to anemia or fatigue. This is important for patient counseling: most common side effect is nausea (often requiring prophylactic antiemetics for the first few months).

Future directions in guidelines: If ongoing trials like those combining olaparib with immunotherapy (KEYLYNK) or targeting HRD (e.g., olaparib in PALB2 mutants, etc.) are positive, guidelines might expand PARPi use beyond strictly BRCA-mutated cases. Also, if a reliable HRD test for breast cancer is validated, one could envision recommendations for HRD+ patients to get PARP inhibitors even without BRCA mutations (similar to ovarian cancer practice). But as of 2025, that's not standard.

In summary, **olaparib is firmly established in guidelines for BRCA-mutant breast cancer:** a standard option in metastatic setting and part of the curative regimen in high-risk early disease. Careful patient selection (germline testing) and attention to prior therapies (no PARP after recent platinum progression) are emphasized. Guidelines also stress the multidisciplinary aspect: genetic counseling for patients and their families given the hereditary implication of a BRCA-based therapy decision.

Additional Relevant Mechanistic/Clinical Notes

- **Quality of Life:** Patients on olaparib generally have better or maintained QoL compared to chemotherapy. OlympiAD reported that global QoL was *not worse* on olaparib than on chemo (actually some metrics favored olaparib) ¹¹³. This supports the idea that for BRCA patients, using oral PARP inhibitors can delay or spare some chemo and its side effects, without compromising well-being. Adjuvant olaparib also didn't show a QoL decrement vs placebo in OlympiA, which is notable given a year of therapy ¹¹³. Fatigue and anemia need monitoring, but manageable.
- **Cross-Resistance with Platinum:** There is a known overlap between sensitivity to platinum chemo and PARP inhibitors because both exploit HRD. Tumors that progress on platinum (especially if progression occurs quickly) often have mechanisms like BRCA reversion already, implying the PARP inhibitor may not work well either. In OlympiAD, patients who had prior platinum within 12 months were excluded, and those with previous progression on platinum did not seem to benefit as much. Conversely, tumors that respond to platinum typically also respond to PARPi. This has led to a debate: in BRCA-mutant metastatic TNBC, should one use a platinum or a PARP inhibitor first? Some retrospective analyses suggest similar efficacy; PARP inhibitors have the convenience advantage. Current practice is to use one or the other, but not both in sequence if the first failed (due to cross-resistance). However, if someone had a platinum years ago in early disease, they could still get a PARP inhibitor at metastasis.
- **Brain Metastases:** Olaparib, being a small molecule, has some ability to penetrate the blood-brain barrier (BBB). There's interest in its role for BRCA-mutant breast cancer patients with brain metastases. Case reports and small series show occasional responses of BRCA mutant brain mets to olaparib, although efficacy is not well established. A phase II trial (ABRAZO cohort in talazoparib) showed intracranial responses in a few patients. Olaparib is not specifically indicated for brain mets, but clinicians may consider it for BRCA patients with stable brain mets or as systemic therapy that might help CNS too. Research is ongoing (e.g., Olaparib in HER2+ with brain mets and a BRCA mutation).
- **Other PARP Inhibitors:** Talazoparib is an alternative PARPi approved in the same indications as olaparib (based on EMBRACA trial). It's slightly more potent in PARP-trapping but has similar efficacy and toxicity profile (talazoparib tends to cause a bit more thrombocytopenia). In practice, many guidelines or pathways treat them interchangeably. Rucaparib and Niraparib are not approved in breast cancer, but niraparib is being tested in some contexts (e.g., BRAVO trial which was halted). If patients are intolerant to one PARPi, occasionally switching to another has been tried, but cross-intolerance is common due to class effects (except maybe different risk of anemia vs liver enzyme elevation, etc.).
- **Combination Therapy Challenges:** Combining olaparib with chemo (like carboplatin) had limited success due to overlapping myelosuppression; required dose reductions that made it not clearly superior to chemo alone. So, despite synergy in theory, in practice sequential use is favored. Combinations with immune checkpoint inhibitors and targeted agents (AKT, MEK inhibitors) are under study and appear more feasible in terms of toxicity, though careful scheduling/dosing is needed.

- **Biomarker Testing:** Germline BRCA testing has become routine for patients with TNBC or with strong family history, largely as a result of the therapeutic implications (PARPi and also implications for platinum use and family counseling). Somatic BRCA testing and broader sequencing are also done for metastatic cases; if a somatic BRCA is found, insurance approvals for olaparib can sometimes be obtained even though label is germline-only – many oncologists would consider a somatic BRCA patient for PARPi off-label or via trial. The companion diagnostic approved is the Myriad BRACAnalysis CDx for germline BRCA ¹¹⁴, which was simultaneously approved with the drug. No official companion diagnostic for HRD in breast yet.
- **Emerging Resistance Mitigations:** To combat resistance, multiple strategies are being explored: **Sequential therapy** (e.g., alternating PARPi with other agents to prevent outgrowth of HR-restored clones), **combination therapy** (as detailed) to suppress pathways like ATR or PI3K that lead to resistance, and even **next-gen PARP inhibitors** designed to evade efflux pumps (like pamiparib, which is less P-gp substrate ¹¹⁵). Another interesting area is **POLθ inhibitors**: since POLθ is crucial for alt-EJ which HRD cells use to survive, inhibitors of POLθ might re-sensitize BRCA-reverted or 53BP1-loss cells by removing their last resort repair. Preclinical work shows POLθ inhibition kills HR-restored, 53BP1-null BRCA1 mutant cells (a strategy to target PARPi-resistant cancers) ¹¹⁶. These might eventually be combined with PARPis or used after PARPi failure.
- **Real-world safety note:** Anemia is the most common Grade ≥3 toxicity; some patients require transfusions or dose reduction. Clinicians monitor CBC monthly; if Hb falls <8 g/dL or symptomatic, holding drug and transfusing is indicated, then resume at reduced dose (e.g., 250 or 200 mg BID). Growth factors are not typically used for neutropenia unless severe infections because neutropenia is usually low-grade. There is a slight risk of **secondary leukemia** as mentioned, especially after prolonged use; ongoing monitoring beyond 1 year is recommended if patients stay on drug (in metastatic setting patients often continue until progression which could be years).

All these points paint the picture of olaparib as a **transformative therapy for BRCA-mutated breast cancer**, with ongoing efforts to expand its reach and overcome resistance. It requires multidisciplinary care – medical oncology to manage therapy and side effects, genetic counseling to manage the hereditary aspects, and potentially coordination with clinical trials for combinations to tackle resistance.

Below is a summary table consolidating the pathways identified across the above sections, specifying whether their upregulation or downregulation affects olaparib sensitivity or resistance, along with the biological rationale and supporting references:

【56†download】 Pathway_Evidence_Table_Olaparib_BreastCancer.csv (Downloadable table of pathways)

FINAL OUTPUT SECTION — Pathway Evidence Table

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Pathway (ID/Name)	Regulation	Effect of Olaparib	Biological Rationale	Reference
REACTOME_HOMOLOGOUS_RECOMBINATION (HRR)	Down	Sensitive	Loss of HR repair (e.g. BRCA1/2 mutation or silencing) creates HR-deficient cells that cannot fix PARP inhibitor-induced DNA breaks, leading to synthetic lethality. ⁶ ⁸	⁶ ⁸
REACTOME_HOMOLOGOUS_RECOMBINATION (HRR)	Up	Resistant	Restoration of HR repair (through BRCA reversion mutations or increased RAD51 function) enables efficient repair of double-strand breaks, allowing tumor cells to survive PARP inhibition. ²⁸ ⁴¹	²⁸ ⁴¹
REACTOME_NONHOMOLOGOUS_END_JOINING_NHEJ	Down	Resistant	Downregulation or loss of 53BP1/RIF1/Shieldin (NHEJ pathway) permits DNA end resection and partial homologous recombination recovery even without BRCA1, thereby bypassing olaparib's lethal effect in BRCA1-deficient cells. ⁴¹ ²⁸	⁴¹ ²⁸

Pathway (ID/Name)	Regulation	Effect of Olaparib	Biological Rationale	Referen
HALLMARK_PI3K_AKT_MTOR_SIGNALING	Up	Resistant	Hyperactivation of PI3K/AKT survival signaling promotes DNA damage repair and cell survival; PARP inhibition induces cytoprotective AKT pathway activity that counteracts cell death (e.g. via BRCA1/2 upregulation and apoptosis suppression). ³²	³² ³¹
HALLMARK_PI3K_AKT_MTOR_SIGNALING	Down	Sensitive	Inhibition or downregulation of PI3K/AKT disrupts compensatory survival pathways and induces "BRCAness" (e.g. reduced BRCA1/2 expression), making cancer cells more reliant on PARP and thus more sensitive to olaparib-induced DNA damage. ³¹	³¹ ³³

Pathway (ID/Name)	Regulation	Effect of Olaparib	Biological Rationale	Referen
HALLMARK_INTERFERON_ALPHA_RESPONSE (STING-mediated)	Up	Sensitive	PARP inhibitor-triggered DNA damage activates the cGAS-STING pathway, leading to Type I interferon production and T-cell recruitment. An inflamed tumor microenvironment with upregulated interferon signaling enhances anti-tumor immunity and improves olaparib response (especially in BRCA-mutant TNBC). ^{24 55}	^{24 55}
REACTOME_ABCFAMILY_PROTEINS_MEDIATED_TRANSPORT (Drug efflux)	Up	Resistant	Overexpression of ATP-binding cassette transporters (e.g. ABCB1/P-gp, ABCG2) increases efflux of olaparib out of cancer cells, lowering intracellular drug levels. This upregulated MDR pathway causes cross-resistance and reduced efficacy of olaparib. ^{39 40}	^{39 40}

Pathway (ID/Name)	Regulation	Effect of Olaparib	Biological Rationale	Reference
ATR/CHK1 Checkpoint Signaling	Up	Resistant	Enhanced ATR-CHK1 activation in response to replication stress allows tumor cells to arrest and repair olaparib-induced DNA damage. Upregulation of this checkpoint pathway stabilizes stalled forks and prevents lethal replication catastrophe, conferring resistance. ⁶⁰ ²⁷	⁶⁰ ²⁷
ATR/CHK1 Checkpoint Signaling	Down	Sensitive	Loss or inhibition of ATR/CHK1 signaling removes the cell's ability to pause and repair DNA damage, leading to unabated replication fork collapse and mitotic death. Downregulating this pathway (e.g. with an ATR or WEE1 inhibitor) magnifies olaparib's cytotoxicity and overcomes resistance. ⁶¹	⁶¹

Pathway (ID/Name)	Regulation	Effect of Olaparib	Biological Rationale	Reference
HALLMARK_APOPTOSIS / p53 Pathway	Up	Sensitive	Upregulation of pro-apoptotic pathways (e.g. functional p53 activity, increased BAX/PUMA) drives prompt cell death in response to DNA damage. Cells with intact or heightened apoptosis signaling undergo apoptosis more readily after PARP inhibition, improving drug efficacy. ⁶ ⁵⁶	⁶ ⁵⁶
TP53 Pathway (G1/S checkpoint, apoptosis)	Down	Resistant	Loss of p53 function or downregulation of apoptotic triggers allows cells to tolerate DNA damage without undergoing cell death. This pathway suppression (common in TNBC) enables survival and accumulation of mutations (e.g. HR restoration), leading to olaparib resistance despite genomic damage. ⁴⁵ ⁹⁴	⁴⁵ ⁹⁴

Pathway (ID/Name)	Regulation	Effect of Olaparib	Biological Rationale	Reference
Fanconi Anemia / DNA Crosslink Repair	Down	Sensitive	Downregulation of Fanconi anemia pathway genes (e.g. FANCD2, FANCA) impairs the repair of DNA interstrand crosslinks and associated HR processes. FA pathway-deficient tumors exhibit HRD and are highly sensitive to PARP inhibitors (similar to BRCA mutations). ²⁰ ²¹	²⁰ ²¹
53BP1/Shieldin-Mediated NHEJ	Down	Resistant	(Same as NHEJ entry above) Loss of 53BP1 or Shieldin complex downregulates error-prone NHEJ and lifts the block on end resection, effectively restoring HR repair in BRCA1-null cells. This adaptive pathway loss undermines olaparib's synthetic lethality and drives resistance. ⁴¹ ²⁸	⁴¹ ²⁸

Pathway (ID/Name)	Regulation	Effect of Olaparib	Biological Rationale	Referen
SLFN11-Dependent DNA Damage Response	Down	Resistant	Silencing of SLFN11 (a protein that links DNA damage to replication arrest) enables cancer cells to continue replicating despite PARP-induced DNA lesions. This downregulation allows tolerance of damage and is associated with intrinsic resistance (notably, SLFN11-low tumors have poorer responses to PARPi). ²⁹	²⁹

Pathway (ID/Name)	Regulation	Effect of Olaparib	Biological Rationale	Reference
PARP1-BER Pathway (Base excision repair)	Down	Resistant	Complete loss or mutation of PARP1 itself, or related BER factors, can make cells less susceptible to PARP-trapping cytotoxicity. If the tumor downregulates the target pathway (e.g. via a PARP1-inactivating mutation), olaparib has reduced effect – the cancer survives due to lack of PARP trapping, albeit at the cost of some DNA repair capacity. ⁴⁵	⁴⁵

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